2019

Total Marks: 70 Time: 3 Hours

(Answer any 5(Five) of the following Questions)

1. a) Define limit and continuity of a function? 3

   b) Solve: $(i)\lim_{x\to 0}[\frac{1}{x^2}-\frac{1}{\sin^2 x}]$ $(ii)\lim_{x\to 0}\frac{e^x-e^{-x}+2\sin x-4x}{x^3}$ by using L' hospitals rule. 3+3

   c) Test the differentiability of the following function at x=0 and x=1: 5

$$f(x) = \begin{cases} -x & when \quad -2 \le x < 0 \\ x & when \quad 0 < x < 1 \\ 3-x & when \quad 1 \le x \le 2 \end{cases}$$

2. a) Find the maxima and minima of the function $f(x) = x^5 - 5x^4 + 5x^3 - 1$ 4

   b) State and prove L'Hospital's rule for indeterminate form. 1+3

   c) When the area of a rectangular with 100m perimeters will be maximum. 6

3. a) Find the area enclosed by the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ 5

   b) Find the area included between the cycloid $x = a(\theta - sin\theta)$, $y = a(1 - cos\theta)$ and 6
   7
   it's base.

   c) Find $(fogoh)(x)$ if $f(x) = \sqrt{x}, g(x) = \frac{1}{x}$ and $h(x) = x^3$ 3

4. a) Prove that 2

$$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$

   b) Evaluate 4 x 3

$$(i)\int_0^{\frac{\pi}{2}} \frac{\sqrt{tanx}}{1+\sqrt{tanx}}dx \qquad (ii)\int_0^1 x^3\sqrt{1+3x^4}\,dx$$

$$(iii)\int_0^1 \frac{ln(1+x)}{1+x^2}\,dx \qquad (iv)\int_0^a \sqrt{\frac{a+x}{a-x}}\,dx$$

5. a) Find the area of the loop of the curve, $r = a\cos 2\theta$ Also, find the area all loops of the curve. 5

   b) Show that 3

$$\int_0^{\frac{\pi}{2}} sin^6\theta\, cos^3\theta\, d\theta = \frac{2}{63}$$

   c) Evaluate (any two): 3 + 3

   i) $\int \frac{dx}{(1-x)\sqrt{1+x}}$ ii) $\int \frac{dx}{\sqrt{1-x^2}\sqrt{sin^{-1}x}}$

   iii) $\int \frac{x^2+1}{(x-1)(x-2)(x+2)}dx$ iv) $\int x\ln x\,dx$

6. a) Prove that

$$\int_a^b f(x)dx = \int_a^b f(z)dz$$

b) Evaluate

$$(i) \int \frac{dx}{\cos^2 x \sin^2 x} \qquad (ii) \int \frac{1-\sin x}{x+\cos x} dx$$

$$(iii) \int \sqrt{1+\sec x}\, dx \qquad (iv) \int \frac{\ln(x+1)}{\sqrt{x+1}} dx$$

7. a) Prove that

$$\beta(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

b) Find the equation of the tangent and normal at the point (x, y) of the curve

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = b^{\frac{2}{3}}$$

1.a

Limit and Continuity

limit of functions: A function $f(x)$ is said to be $\boxed{\text{Converged}}$ to a limit at $x=a$ if for any $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - l| < \epsilon$ whenever $\boxed{a-\delta < x < a+\delta}$ but $x \neq a$, then we can write,

$$\underset{x \to a}{Lt}\ f(x) = l$$

$$\text{or } \lim_{x \to a} f(x) = l$$

There are two kinds of limit such that:

① Left hand limit $\rightarrow \lim_{x \to a^-} f(x) = l_1$

② Right hand limit $\rightarrow \lim_{x \to a^+} f(x) = l_2$

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Continuity of a function:

A function $f(x)$ is said to be continuous at $x=a$ if it exists and equal to $f(a)$.

In other words,

$f(x)$ is said to be a continuous function at $x=a$ if,

$$\boxed{\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)}$$

(b)

i) $\lim_{x \to 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$

Given,

$$\lim_{x \to 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right] \quad [\infty - \infty]$$

$$= \lim_{x \to 0} \left[\frac{\sin^2 x - x^2}{x^2 \sin^2 x} \right] \quad [0/0]$$

$$= \lim_{x \to 0} \left(\frac{2 \sin x \cos x - 2x}{x^2\, 2 \sin x \cos x + 2 \sin^2 x\, x} \right)$$

$$= \lim_{x \to 0} \left(\frac{\sin 2x - 2x}{x^2 \sin 2x + 2x \sin^2 x} \right) \quad [0/0]$$

$$= \lim_{x \to 0} \left(\frac{2 \cos 2x - 2}{x^2 . 2\cos 2x + 2x \sin 2x + 2\sin^2 x + 2x . 2 \sin x \cos x} \right)$$

$$= \lim_{x \to 0} \left[\frac{2\cos 2x - 2}{2x^2 \cos 2x + 2x \sin 2x + 2\sin^2 x + 2x \sin 2x} \right]$$

$$\lim_{x \to 0} \left[\frac{2\cos 2x - 2}{2x^2 \cos 2x + 4x \sin 2x + 2\sin^2 x} \right]$$

$$= \lim_{x \to 0} \left[\frac{\cancel{2}(\cos 2x - 1)}{\cancel{2}(x^2 \cos 2x + 2x \sin 2x + \sin^2 x} \right]$$

$$= \lim_{x \to 0} \left[\frac{-2\sin 2x}{-2x^2 \sin 2x + 2x\cos 2x + 2\sin x \cos x + 4x \cos 2x + 2\sin 2x} \right]$$

$$= \lim_{x \to 0} \left[\frac{-2\sin 2x}{-2x^2 \sin 2x + 6x \cos 2x + \cancel{1} \sin x 2x + 2\sin 2} \right]$$

$$= \lim_{x \to 0} \left[\frac{-2\sin 2x}{-2x^2 \sin 2x + 6x \cos 2x + 3\sin 2x} \right]$$

$$= \lim_{x \to 0} \left[\frac{-4\cos 2x}{-4x\sin 2x + 4x^2 \cos 2x + 6\cos 2x - 12x \sin 2x + 6\cos 2x} \right]$$

$$= \lim_{x \to 0} \frac{-4\cos 2x}{-0+0+6-0+6} = \frac{-\cancel{4}}{\cancel{12}\,3} = -\frac{1}{3}$$

(Ans)

ii)

$$\lim_{x \to 0} \frac{e^x - e^x + 2\sin x - 4x}{x^3}$$

$$= \lim_{x \to 0} \frac{e^x - e^x + 2\cos x - 4}{3x^2} \quad [0/0]$$

$$= \lim_{x \to 0} \frac{e^x - e^x - 2\sin x}{6x} \quad [0/0]$$

$$= \lim_{x \to 0} \frac{e^x - e^x - 2\cos x}{6}$$

$$= \frac{1 - 1 - 2}{6} = \frac{-\cancel{2}}{\cancel{6}\,3}$$

$$= -\frac{1}{3}$$

Prob: ... tion $f(x)$ is defined ...

(c)

given,

$$f(x) = \begin{cases} -x & \text{when } -2 \le x < 0 \\ x & \text{when } 0 < x < 1 \\ 3-x & \text{when } 1 \le x \le 2 \end{cases}$$

at $x = 0$

when $x > 0$ then $f(x) = x$

$$Rf'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{1}{h} (0 + h - 0)$$

$$= \frac{h}{h}$$

$$= \underline{1}$$

when $x < 0$ then $f(x) = -x$

$$Lf'(0) = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} -\frac{1}{h} (0 - h - 0)$$

$$= \frac{-h}{-h}$$

$$= \underline{1}$$

Hence, we have $Rf'(0) = Lf'(0)$.

Therefore, the function $f(x)$ is differentiable at $x = 0$.

again at $x = 1$.

if $x > 1$ then $f(x) = 3 - x$

$$Rf'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{3 - (1+h) - (3-1)}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{3} - \cancel{1} - h - \cancel{3} + \cancel{1}}{h}$$

$$= \lim_{h \to 0} \frac{-h}{h}$$

$$= -1$$

if $x < 1$ then $f(x) = x$

$$Lf'(1) = \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \to 0} \frac{1 - h - 1}{-h}$$

$$= \lim_{h \to 0} \frac{-h}{-h}$$

$= \underline{1}$.

Hence we have $Rf'(1) \neq Lf'(1)$.

Therefore, the function is not differentiable at $x = \underline{1}$.

(A).

2. (a) Given,

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

$$\therefore f'(x) = 5x^4 - 20x^3 + 15x^2 - 0$$

$$= 5x^4 - 20x^3 + 15x^2 \quad \text{——— (i)}$$

For a maximum or a minimum values of $f(x)$, $f'(x) = 0$.

i.e; $5x^4 - 20x^3 + 15x^2 = 0$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow x^2 - 3x - x + 3 = 0$$

$$\Rightarrow x(x-3) - 1(x-3) = 0$$

$$\Rightarrow (x-3)(x-1) = 0$$

$$\therefore x = 1, 3$$

Again, $f''(x) = 20x^3 - 60x^2 + 30x$

when, $x = 1$, then,

$$f''(1) = 20 - 60 + 30 = -10 \ (-ve)$$

$f(x)$ is maximum when $x = 1$ and the max value is $f(1) = 1 - 5 + 5 - 1$

$$= 0 \quad \text{(Ans:)}$$

when, $x=3$, then,

$$f''(3) = 20(3)^3 - 60(3)^2 + 30\times 3$$
$$= 540 - 540 + 90$$
$$= 90 \ (+ve)$$

$\therefore f(x)$ is minimum when $x=3$ and the minimum value is $f(3) = -28$ (Ans:)

---o---

2. (b) L' Hospital's Rule:

statement: If $\phi(x)$, $\psi(x)$ as also their derivatives.

$\phi(x)$ and $\psi(x)$ are continuous at $x=a$ and if $\phi(a) = \psi(a) = 0$.

$\left[\text{i.e; } \lim_{x\to a} \phi(x) = \lim_{x\to a} \psi(a) = 0\right]$ then,

$$\lim_{x\to a} \frac{\phi(x)}{\psi(x)} = \lim_{x\to a} \frac{\phi'(x)}{\psi'(x)} = \frac{\phi'(a)}{\psi'(a)} = \psi'(a) \neq 0$$

Example: Evaluate $\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$ by L' Hospitals rule.

Solve: Given,

$$\lim_{x\to 0} \frac{e^{x} - e^{-x} - 2x}{x - \sin x} \quad \left[\frac{0}{0} \text{ form}\right]$$

$$= \lim_{x\to 0} \frac{e^{x} + e^{-x} - 2}{1 - \cos x} \quad \left[\frac{0}{0} \text{ form}\right]$$

$$= \lim_{x\to 0} \frac{e^{x} - e^{-x} - 0}{+\sin x} \quad \left[\frac{0}{0} \text{ form}\right]$$

$$= \lim_{x\to 0} \frac{e^{x} + e^{-x}}{\cos x}$$

$$= \frac{e^{0} + e^{-0}}{\cos 0} = \frac{1+1}{1} = 2$$ (Ans:)

——— o ———

C

we know that the perimetere of rectangle $s = 2(x+y)$

$\therefore y = (s-2x)/2$

the area of a rectangle,

$$A = xy$$
$$= x \cdot (s-2x)/2$$
$$= x/2 \cdot (s-2x)$$
$$= 1/2 \cdot (sx - 2x^2)$$

$$\therefore \frac{dA}{dx} = 1/2\,(s-4x)$$

$$\frac{dA}{dx} = 0$$
$$\Rightarrow 1/2(s-4x) = 0$$
$$\Rightarrow s - 4x = 0$$
$$\Rightarrow x = s/4$$
$$\therefore y = s/4$$

we have, $s = 100m$

$\therefore x = 25 \quad y = 25.$

Again,

$$d^2A/dx^2 = -4$$
$$= -ve$$

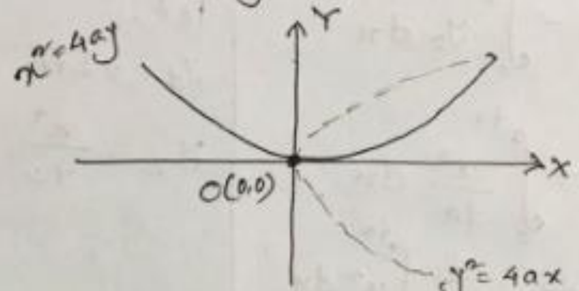
Hence, the area of the rectangle will be maximum when the rectangle is a square.

(A).

2012 Math

03

ⓐ Area enclosed by the parabola $y^2 = 4ax$ and $x^2 = 4ay$



Given equations,

$$y^2 = 4ax \text{ --- (I)} \Rightarrow x = \frac{y^2}{4a}$$
$$x^2 = 4ay \text{ --- (II)}$$

$$\Rightarrow \left(\frac{y^2}{4a}\right) = 4ay$$

$$\Rightarrow \frac{y^4}{16a^2} = 4ay$$

$$\Rightarrow y^4 = 64a^3y$$

$$\Rightarrow y = \sqrt[3]{64a^3} = 4a$$

or, $y = 0$; [the curve touches the origin]

~~∴ y = (0, 4a 0 and 4a~~

From (I),

if $y = 0$ then $x = 0$

if $y = 4a$ then $x = 4a$

$\therefore$ The intersection points of (I) and (II) are $O(0,0)$ and $A(4a,4a)$

$\therefore$ The area of the region OPAQ is,

$$\int_0^{4a} y_1\,dx - \int_0^{4a} y_2\,dx$$

let, $y_1 = 2\sqrt{ax}$.

$y_2 = \frac{x^2}{4a}$

$$= \int_0^{4a} 2\sqrt{ax} - \int_0^{4a} \frac{x^2}{4a}\,dx$$

$$= 2\sqrt{a}\int_0^{4a} \sqrt{x}\,dx - \frac{1}{4a}\int_0^{4a} x^2\,dx$$

$$= 2\sqrt{a}\left[\frac{x^{3/2}}{\frac{3}{2}}\right]_0^{4a} - \frac{1}{4a}\left[\frac{x^3}{3}\right]_0^{4a}$$

$$= 2\sqrt{a}\left[\frac{(4a)^{3/2}}{3/2} - 0\right] - \frac{1}{4a}\left[\frac{(4a)^3}{3} - 0\right]$$

$$= 2\sqrt{a}\;\frac{8a^{3/2}}{3/2} - \frac{64a^3}{4a\times 3}$$

$$= \frac{16a^{\frac{3}{2}+\frac{1}{2}}}{3/2} - \frac{64a^2}{12}$$

$$= \frac{16a^2}{1}\times\frac{2}{3} - \frac{16a^2}{3}$$

$$= \frac{32a^2}{3} - \frac{16a^2}{3}$$

$$= \boxed{\frac{16a^2}{3}}$$

(Ans.)

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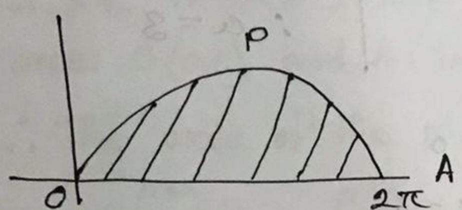
Example-03

2019 3(b)

Find the area of the cycloid $x = a(\theta - \sin\theta)$ and $y = a(1-\cos\theta)$ bounded by the x-axis.



$\Rightarrow$ The equations are:

$$x = a(\theta - \sin\theta),\ y = a(1-\cos\theta)$$

$$\therefore \frac{dx}{d\theta} = a(1-\cos\theta)$$

$\therefore$ The required area of the region = Area of OAPO.

$$= \int_0^{2\pi} y \frac{dx}{d\theta} \cdot d\theta$$

$$= \int_0^{2\pi} a(1-\cos\theta)(1-\cos\theta) \cdot d\theta$$

$$= \int_0^{2\pi} a^2(1-\cos\theta)^2 \, d\theta$$

$$= a^2 \int_0^{2\pi} \left(2\sin^2\frac{\theta}{2}\right)^2 d\theta$$

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$$= 4a^2 \int_0^{2\pi} \sin^4 \frac{\theta}{2} \, d\theta$$

$$= 4a^2 \int_0^{2\pi} \sin^4\phi \, 2 \, d\phi$$

$$= 8a^2 \int_0^{\pi} \sin^4\phi \cdot \cos^0\phi \, d\phi$$

$$= 2 \cdot 8a^2 \int_0^{\pi/2} \frac{\Gamma\frac{4+1}{2} \, \Gamma\frac{1}{2}}{2\,\Gamma\frac{4+0+2}{2}}$$

$$= 16a^2 \frac{\Gamma\frac{5}{2} \cdot \sqrt{\pi}}{2\,\Gamma 3}$$

$$= 16a^2 \frac{\Gamma 1+\frac{3}{2} \, \sqrt{\pi}}{2 \times 2}$$

$$\Gamma\frac{5}{2} = \Gamma 1+\frac{3}{2}$$

$$= \frac{3}{2}\,\Gamma\frac{3}{2}$$

$$= \frac{3}{2}\,\Gamma 1+\frac{1}{2}$$

$$= \frac{3}{2} \cdot \frac{1}{2}\,\Gamma\frac{1}{2}$$

$$= \frac{3}{4}\,\Gamma\pi$$

Let,

$$\frac{\theta}{2} = \phi$$

$$\Rightarrow \frac{1}{2} = \frac{d\phi}{d\theta} \quad \boxed{\therefore \theta = 2d\phi}$$

θ	0	2π
ϕ	0	π

$$= 16a^2 \frac{\frac{3}{4}\sqrt{\pi} \cdot \sqrt{\pi}}{4}$$

$$= 4a^2 \times \frac{3}{4} \times \pi$$

$$= 3\pi a^2$$

2012 Math

3 (c) Given, $f(x) = \sqrt{x}$, $g(x) = \frac{1}{x}$ and $h(x) = x^3$

$h(x) = x^3$

$$g \circ h(x) = \left(\frac{1}{x}\right)^3 = \frac{1}{x^3}$$

$$\therefore f \circ g \circ h(x) = \frac{1}{(\sqrt{x})^3}$$

$$= \frac{1}{(x^{\frac{1}{2}})^3}$$

$$= \frac{1}{x^{3/2}}$$

4. (a)

a) Here, $\int_0^a f(x)dx = \int_a^0 f(a-t)(-dt)$

$= -\int_a^0 f(a-t)\, dt$

$= \int_0^a f(a-t)\, dt$

$= \int_0^a f(a-x)dx$

putting

$x = a-t$

$\Rightarrow 1 = -dt/dx$

$\Rightarrow dx = -dt$

$-\int_0^a f(u)du = \int_a^0 f(v)dv$

$\int_a^b f(u)du = \int_a^b f(v)dv$

4 (b)

b)(i) Here, $\displaystyle\int_0^{\pi/2} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\, dx$

Let,

$$I = \int_0^{\pi/2} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\sqrt{\sin x}}{\sqrt{\cos x}}}{1+\frac{\sqrt{\sin x}}{\sqrt{\cos x}}} \cdot dx$$

$$= \int_0^{\pi/2} \frac{\frac{\sqrt{\sin x}}{\sqrt{\cos x}}}{\frac{\sqrt{\cos x}+\sqrt{\sin x}}{\sqrt{\cos x}}} \cdot dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}\cdot\sqrt{\cos x}\cdot dx}{\sqrt{\cos x}\left(\sqrt{\cos x}+\sqrt{\sin x}\right)}$$

$$\therefore I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}}\, dx \; --- \; (1)$$

Again,

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}+\sqrt{\sin\left(\frac{\pi}{2}-x\right)}} \cdot dx$$

$\because \int_b^a f(x)\,dx = \int_b^a f(a-x)\,dx$

$$\therefore I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}} \quad \text{--- (II)}$$

Adding (I) + (II)

$$2I = \int_0^{\pi/2} \left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\sin x}+\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}} \right) dx$$

$$= \int_0^{\pi/2} dx \qquad = \left[x \right]_0^{\pi/2} \qquad = \frac{\pi}{2} - 0$$

$$\Rightarrow 2I = \frac{\pi}{2}$$

$$\Rightarrow I = \frac{\pi}{4}$$

$$\therefore \int_0^{\pi/2} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}} = \frac{\pi}{4}$$

f

4.(ii)

$$\int_0^1 x^3\sqrt{1+3x^4}\,dx$$

$$I = \int_0^1 x^3\sqrt{1+3x^4}\,dx$$

<u>Putting</u>

$$1+3x^4 = t^2.$$

$$12x^3\,dx = 2t\,dt$$

$$x^3\,dx = \tfrac{1}{6}\,t\,dt$$

$$x=0 \rightarrow t=1$$

$$x=1 \rightarrow t=2$$

$$I = \int_1^2 \frac{1}{6}\,t\cdot t\,dt$$

$$= \tfrac{1}{6}\int_1^2 t^2\,dt$$

$$= \tfrac{1}{6}\left[\frac{t^3}{3}\right]_1^2$$

$$= \frac{1}{6}\left[\frac{8}{3}-\frac{1}{3}\right]$$

$$= \frac{1}{6}\left[\frac{7}{3}\right] = \frac{7}{18}$$

iii) $\int_0^1 \frac{\ln(1+x)}{1+x^2} = I$

$\Rightarrow \int_0^{\pi/4} \frac{\ln(1+\tan\theta)}{1+\tan^2\theta} \cdot \sec^2\theta \, d\theta$

$\Rightarrow \int_0^{\pi/4} \ln(1+\tan\theta) \, d\theta$

$\Rightarrow \int_0^{\pi/4} \ln\left[1+\tan\left(\pi/4-\theta\right)\right) d\theta$

$= \int_0^{\pi/4} \ln\left[1+\frac{1-\tan\theta}{1+\tan\theta}\right] d\theta$

$= \int_0^{\pi/4} \ln\left(\frac{1+\tan\theta+1-\tan\theta}{1+\tan\theta}\right) d\theta$

$= \int_0^{\pi/4} \ln\left(\frac{2}{1+\tan\theta}\right) d\theta$

$x = \tan\theta$

$dx = \sec^2\theta \, d\theta$

$x \to 0, \theta \to 0$

$x \to 1, \theta = \pi/4$

$\sec^2\theta = 1+\tan^2\theta$

$\int_0^a f(a-x)\,dx = \int_0^a f(x)\,dx$

$$= \int_0^{\pi/4} \left[\ln(2) - \ln(1+\tan\theta) \right] d\theta$$

$$= \ln(2) \int_0^{\pi/4} d\theta - \underbrace{\int_0^{\pi/4} \ln(1+\tan\theta)\, d\theta}_{I}$$

$$2I = \ln(2) \cdot [\theta]_0^{\pi/4}$$

$$= \ln(2) \; \pi/4$$

$$I = \pi/8 \; \ln(2)$$ Answer.

(IV)

$$I = \int_0^a \sqrt{\frac{a+x}{a-x}}\, dx$$

$$= \int_0^a \sqrt{\frac{(a+x)(a+x)}{(a-x)(a+x)}}\, dx$$

$$= \int_0^a \frac{a+x}{\sqrt{a^2-x^2}}\, dx$$

$$= \int_0^a \frac{a}{\sqrt{a^2-x^2}}\, dx + \int_0^a \frac{x}{\sqrt{a^2-x^2}}\, dx$$

$$= a\int_0^a \frac{dx}{\sqrt{a^2-x^2}} + \frac{1}{(-2)} \int_0^a \frac{-2x\, dx}{\sqrt{a^2-x^2}}$$

$$= \left[a \sin^{-1}\left(\frac{x}{a}\right)\right]_0^a - \frac{1}{2} \int_0^a \frac{-2x\, dx}{\sqrt{a^2-x^2}}$$

$a^2-x^2 = t$

$-2x\,dx = d$

$$= \left[a \sin^{-1} \frac{x}{a}\right]_0^a - \frac{1}{2} \int t^{-1/2}\, dt$$

$$= \left[a \sin^{-1} \frac{x}{a}\right]_0^a - \frac{3}{2}\, t^{1/2}$$

$$I = \left[a\sin^{-1}\frac{x}{a}\right]_0^a - \left[t^{1/2}\right]_0^a$$

$$= \left[a\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a - \left[\sqrt{a^2-x^2}\right]_0^a$$

$$= \left[a\sin^{-1}\frac{a}{a} - a\sin^{-1}\left(\frac{0}{a}\right)\right] - \left[\sqrt{a^2-a^2} - \sqrt{a^2-0}\right]$$

$$= \frac{\pi}{2}a - 0 - 0 + a$$

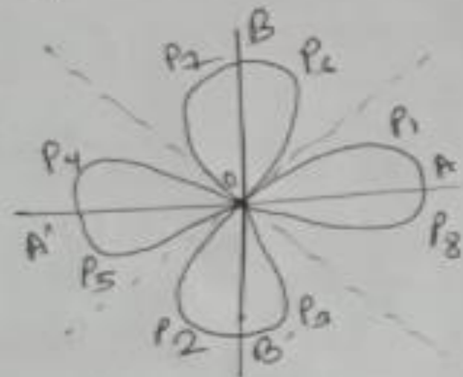
$$= \frac{\pi}{2}a + a$$

Ans

$=$

5

a) Find the area of a loop of the curve $r = a\cos\theta$.



In tracing the curve, we notice that, as θ increase from 0 to $\frac{\pi}{4}$, r diminishes from a to 0. the portion AP_1O being thus traced. As θ increase from $\frac{\pi}{4}$ to $\frac{3\pi}{4}$, r is negative throughout, & the corresponding portion of the curve which is traced is $OP_2B'P_3O$. Then as θ increases from

$\frac{3\pi}{4}$ to $\frac{5\pi}{4}$, r remain positive portion $OP_4A'P_5O$ of the curve is traced. As θ increases from $\frac{5}{4}\pi$ to $\frac{7}{4}\pi$, r is again negative & we get the portion $OP_6B'P_7O$ of the curve. when θ increases from $\frac{7}{4}\pi$ to 2π, r is positive & the portion OP_8AO of the curve is discribed. The curve thus consists of four equal loop as shown in the figure.

area of one loop,

$$= 2 \text{ area } AP_1O$$

$$= 2 \cdot \frac{1}{2} \int_0^{\pi/4} r^2 d\theta$$

$$= a^2 \int_0^{\pi/4} \cos^2 2\theta d\theta$$

$$= \frac{1}{8}\pi a^2$$

Area of all the loops of the curve;

$$4 \times \frac{1}{8}\pi a^2$$

$$\Rightarrow \frac{1}{2}\pi a^2$$

(A).

(b)

$$\int_0^{\pi/2} \sin^6\theta \cos^3 d\theta$$

$$= \frac{\Gamma\frac{6+1}{2}\ \Gamma\frac{3+1}{2}}{2\Gamma\frac{6+3+2}{2}}$$

$$= \frac{\Gamma\frac{7}{2}\ \Gamma 2}{2\Gamma\frac{11}{2}}$$

$$= \frac{\Gamma 1+\frac{5}{2}\ \Gamma 2}{2\Gamma 1+\frac{9}{2}}$$

$$= \frac{\frac{5}{2}\Gamma\frac{5}{2}\ 1!}{2\frac{9}{2}\Gamma\frac{9}{2}}$$

$$= \frac{\frac{5}{2}\Gamma 1+\frac{3}{2}}{9\cdot\Gamma 1+\frac{7}{2}}$$

(i) $\displaystyle\int \frac{dx}{(1-x)\sqrt{1+x}}$

put,
$1+x = z^2$
$\Rightarrow 2z\,dz = dx$
$\Rightarrow z^2 - 1 = x$
$z = \sqrt{x+1}$

$$= \int \frac{2z\,dz}{(1-z^2-1)z}$$

$$= -\int \frac{2dz}{z^2}$$

$$= -2\int z^{-2}\,dz$$

$$= -2\,\frac{z^{-2+1}}{-2+1} + C$$

$$= -2\,\frac{z^{-1}}{-1} + C$$

$$= 2\,\frac{1}{z} + C$$

$$= \frac{2}{\sqrt{x+1}} + C$$

(A).

(c)

(11) $\int \frac{dx}{\sqrt{\sin^{-1}x}\sqrt{1-x^2}}$

put, $z = \sin^{-1}x$

$\Rightarrow dz = \frac{1}{\sqrt{1-x^2}}\,dx$

$= \int \frac{1}{\sqrt{z}}\,dz$

$= 2\sqrt{z} + c$

$= 2\sqrt{\sin^{-1}x} + c$ (A).

5(c) iii

$$\int \frac{x^2+1}{(x-1)(x-2)(x+2)}\,dx$$

$$\Rightarrow \frac{x^2+1}{(x-1)(x-2)(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{x+2}$$

$$\Rightarrow x^2+1 = A(x-2)(x+2) + B(x-1)(x+2) + C(x-1)(x-2)$$

when $x=1$

$$\Rightarrow 2 = A(1-2)(1+2) + B\cdot 0 + C\cdot 0$$

$$\Rightarrow 2 = -3A$$

$$\Rightarrow A = -\frac{2}{3}$$

when $x=2$

$$\Rightarrow 5 = A\cdot 0 + B(2-1)(2+2) + C\cdot 0$$

$$\Rightarrow 5 = 4B$$

$$\Rightarrow B = \frac{5}{4}$$

When, $x=-2$

$$\Rightarrow 5 = A\cdot 0 + B\cdot 0 + C(-2-1)\cdot(-2-2)$$

$\Rightarrow 5 = C(-12)$

$\Rightarrow C = \frac{-5}{12}$

Hence,

$$\int \frac{-2/3 \, dx}{(x-1)} + \int \frac{5/4}{x-2} dx + \int \frac{-5/12}{x+2} dx$$

$$\Rightarrow -2/3 \ln|x-1| + 5/4 \ln|x-2| + (-5/12) \ln(x+2) + c$$

$$\Rightarrow -\frac{2}{3} \ln|x-1| + 5/4 \ln|x-2| - 5/12 \ln|x+2| + c$$

(Ans;)

(iv) $\int x \ln x \, dx$

$= \int \ln x \cdot x \, dx$

$= \ln x \int x \, dx - \int \left[\frac{d}{dx}(\ln x) \int x \, dx \right] dx$

$= \frac{x^2}{2} \ln x - \int \frac{1}{x} \cdot \frac{x^2}{2} \, dx$

$= \frac{x^2}{2} \ln x - \frac{1}{2} \cancel{\int \frac{x^2}{x} dx} \int x \, dx$

$= \frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{x^2}{2} + c$

$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + c$

(A).

6(a)

#

$$\int_a^b f(x)\,dx = \int_a^b f(z)\,dz$$

$$L.H.S = \int_a^b f(x)\,dx$$

$$= \Big[f(x)\Big]_a^b$$

$$= f(b) - f(a)$$

$$R.H.S = \int_a^b f(z)\,dz$$

$$= \Big[f(z)\Big]_a^b$$

$$= f(b) - f(a)$$

$$\therefore L.H.S = R.H.S$$

6) b

i) $\int \frac{dx}{\cos^2 x . \sin^2 x}$

$= \int \frac{(\cos^2 x + \sin^2 x)}{\sin^2 x . \cos^2 x} dx$

$= \int \left(\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x} \right) dx$

$= \int (\operatorname{cosec}^2 x + \sec^2 x) \, dx$

$= \tan x - \cot x + c$ (Ans:)

(ii) $\int \frac{1-\sin x}{x+\cos x} dx$

Putting,

$x + \cos x = \theta$

$\Rightarrow 1 - \sin x = \frac{d\theta}{dx}$

$\Rightarrow (1-\sin x) \, dx = d\theta$

$= \int \frac{d\theta}{\theta}$

$= \log|\theta| + c$

$= \log|x + \cos x| + c$

(Ans:)

$$\int \sqrt{1+secx}\,dx \quad = \int \sqrt{1+\frac{1}{cosx}}\,dx$$

$$= \int \sqrt{\frac{cosx+1}{cosx}}\,dx$$

$$= \int \sqrt{\frac{2cos^2\left(\frac{x}{2}\right)}{1-2sin^2\left(\frac{x}{2}\right)}}\,dx \; = \sqrt{2}\int \frac{cos\left(\frac{x}{2}\right)}{\sqrt{1-2sin^2\left(\frac{x}{2}\right)}}\,dx$$

$$= \sqrt{2}\int \frac{cos\left(\frac{x}{2}\right)}{\sqrt{1-2sin^2\left(\frac{x}{2}\right)}}\,dx$$

$$\sqrt{2}sin\left(\frac{x}{2}\right) = z$$

$$\Rightarrow \sqrt{2}cos\left(\frac{x}{2}\right).\frac{1}{2}.dx = dz$$

$$\Rightarrow cos\left(\frac{x}{2}\right)dx = \sqrt{2}dz$$

$$= \sqrt{2}\int \frac{\sqrt{2}dz}{\sqrt{1-z^2}}$$

$$= 2\;sin^{-1}z + c$$

$$= 2\;sin^{-1}\left(\sqrt{2}sin\left(\frac{x}{2}\right)\right) + c$$

(iii) $\int \sqrt{1+\sec x}\, dx$

(iv) $\int \frac{\ln(x+1)}{\sqrt{x+1}}\, dx$

$= \ln(x+1) \int \frac{dx}{\sqrt{x+1}} - \int \left\{ \frac{d}{dx} \ln(x+1) \int \frac{dx}{\sqrt{x+1}} \right\} dx$

$= \ln(x+1) \cdot 2\sqrt{x+1} - \int \left(\frac{1}{x+1} \times 2\sqrt{x+1} \right) dx$

$= 2\sqrt{x+1}\, \ln(x+1) - 2\int \frac{dx}{\sqrt{x+1}} + c'$

$= 2\sqrt{x+1}\, \ln(x+1) - 2 \cdot 2\sqrt{x+1} + c$

$= 2\sqrt{x+1}\, \ln(x+1) - 4\sqrt{x+1} + c$ (Ans:)

Formula:

$$\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + c$$

7. (a)

Question:- Establish the relation between Gamma function and Beta function.

$\boxed{\text{OR}}$:- Prove that, $\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma (m+n)}$.

Soln:- we know that,

$$\Gamma m = \int_0^\infty e^{-x} \cdot x^{m-1} dx \quad \text{--- (1)}$$

Let, $x = \lambda y$

$$\Rightarrow 1 = \lambda \frac{dy}{dx}$$

$$\therefore dx = \lambda dy$$

x	0	∞
y	0	∞

From (1), we can write,

$$\Gamma m = \int_0^\infty e^{-\lambda y} (\lambda y)^{m-1} \lambda dy.$$

$$\Rightarrow \Gamma m = \int_0^\infty e^{-\lambda y} \lambda^m y^{m-1} dy. \quad \text{--- (2)}$$

Now, we can write,

$$\Gamma n = \int_0^\infty e^{-\lambda} \lambda^{n-1} d\lambda \quad \text{--- (3)}.$$

Now, Multiplying (2) and (3); we get,

$$\Gamma m \, \Gamma n = \int_0^\infty \int_0^\infty e^{-\lambda y - \lambda} \lambda^{m+n-1} y^{m-1} dy \, d\lambda$$

$$\Gamma m \, \Gamma n = \int_0^\infty \left[\int_0^\infty e^{-\lambda(1+y)} \lambda^{m+n-1} \, d\lambda \right] y^{m-1} \, dy.$$

$$= \int_0^\infty \frac{\Gamma(m+n)}{(1+y)^{m+n}} \, y^{m-1} \, dy \quad \left[\int_0^\infty e^{-kt} t^{n-1} \, dt = \frac{\Gamma n}{k^n} \right]$$

$$= \Gamma(m+n) \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}} \, dy.$$

$$= \Gamma(m+n) \; \beta(m,n) \quad \left[\beta(m,n) = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}} \, dy \right]$$

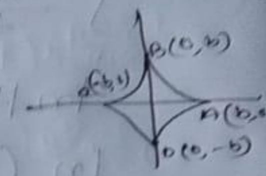
$$\therefore \Gamma m . \Gamma n = \Gamma(m+n) . \beta(m,n)$$

$$\therefore \beta(m,n) = \frac{\Gamma m . \Gamma n}{\Gamma(m+n)} . \text{(Proved)}$$

7.(b)

axes.
If $y=0$ then $^{2/n}$ $^{2/n}$

※ Find the equation of the tangent and normal at the point (x,y) of the curve $x^{2/n}+y^{2/n}=b^{2/n}$.



Soln:-

The curve is $x^{2/n}+y^{2/n}=b^{2/n}$ is symmetrical about both of the ~~area~~ axes,

If $y=0$ then $x^{2/n}=b^{2/n}$

$\therefore x=a$

$\therefore$ The intersection points of x-axis, $A(b,0)$, $C(-b,0)$.

$\therefore$ Required area $=4\times$ area of the region CABO

$\therefore A=4\int_0^b y\,dx$ —— (1)

let,
$a=a\cos^3\theta,\ y=a\sin^3\theta$
$\Rightarrow 1=-3a\cos^2\theta.\sin\theta\frac{d\theta}{dx}$.
$dx=-3a\sin\theta.\cos^2\theta d\theta$.

x	0	a
θ	$\pi/2$	0

From (1), we get

$A=4\int_{\pi/2}^{0} a\sin^3\theta(-3a\cos^2\theta\sin\theta)d\theta$

$=12a^2\int_0^{\pi/2} a\sin^4\theta.\cos^2\theta\,d\theta$.

$=12a^2\ \dfrac{\Gamma\frac{4+1}{2}.\Gamma\frac{2+1}{2}}{2\Gamma\frac{4+2+2}{2}}$

$=12a^2\ \dfrac{\Gamma\frac{5}{2}\ \Gamma\frac{3}{2}}{2\Gamma 4}$

$=12a^2\ \dfrac{\frac{3}{2}.\frac{1}{2}\sqrt{\pi}.\frac{1}{2}\sqrt{\pi}}{2.3!}$

$=a^2\ \frac{3}{8}\ \pi$ (Ans)